CASE: US AIRLINES

Industry Analysis & Porter´s Five Forces

Session 5

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Agenda for today

Please try to answer this **4 questions** :

Question 1 → How did the industry perform in recent years?

Question 2 → To what extent can the industry's low average profitability during this period be attributed to the structure of the industry?

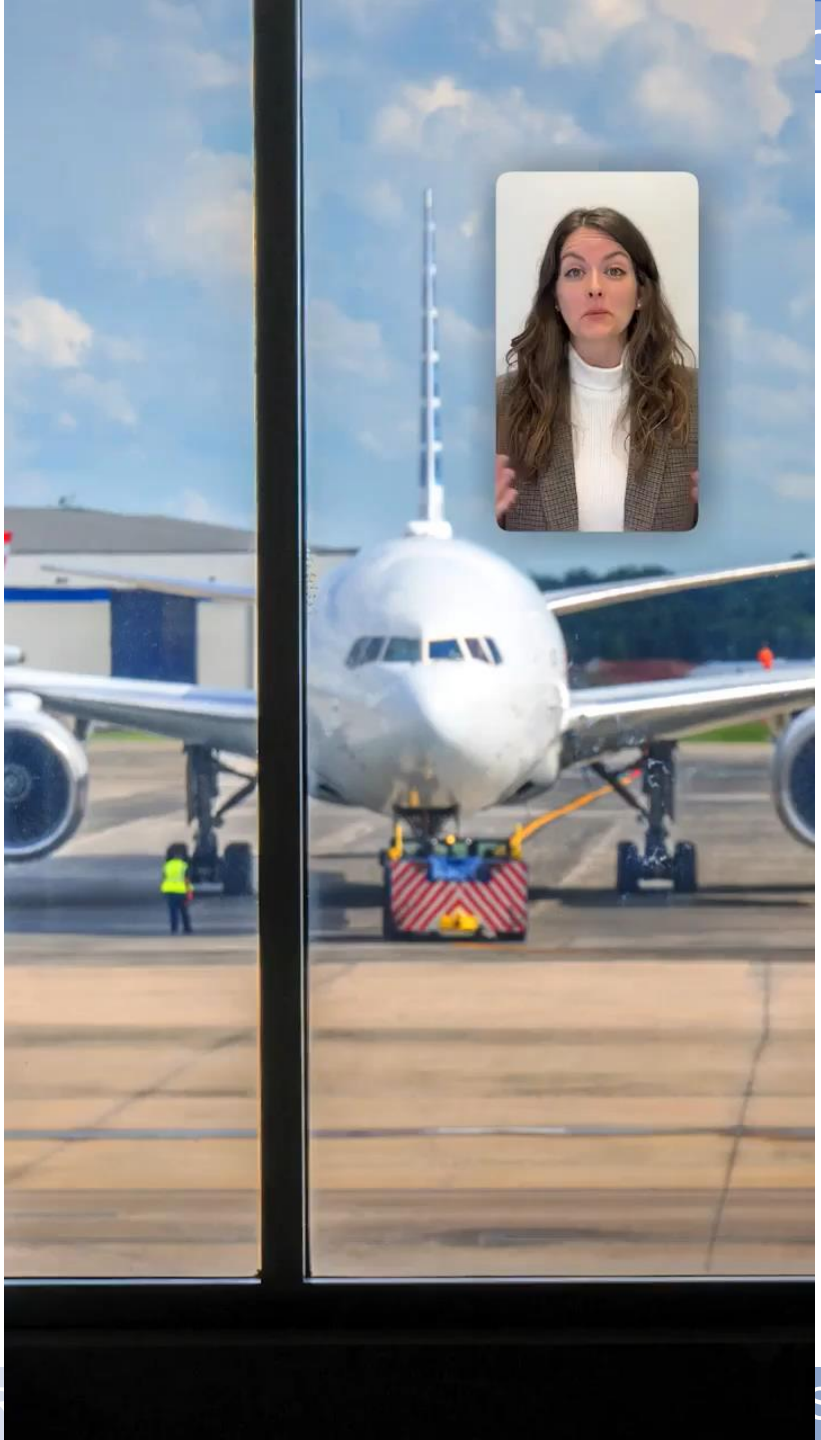
Question 3 → Which of Porter's five forces has had the biggest impact in depressing industry profitability?

Question 4 → How will industry profitability develop during the next five years?

What would you do as manager of an airline?

2024 Airline Industry

(Wall Street Journal)



The Global Airline Industry : the market [Kathryn McFarland] 3 mts.
https://www.youtube.com/watch?v=tmhMLVHJIEs



Question 1→

How did the industry perform in recent years?

Q1: How did the industry perform in recent years?

General characteristics of US Airlines industry

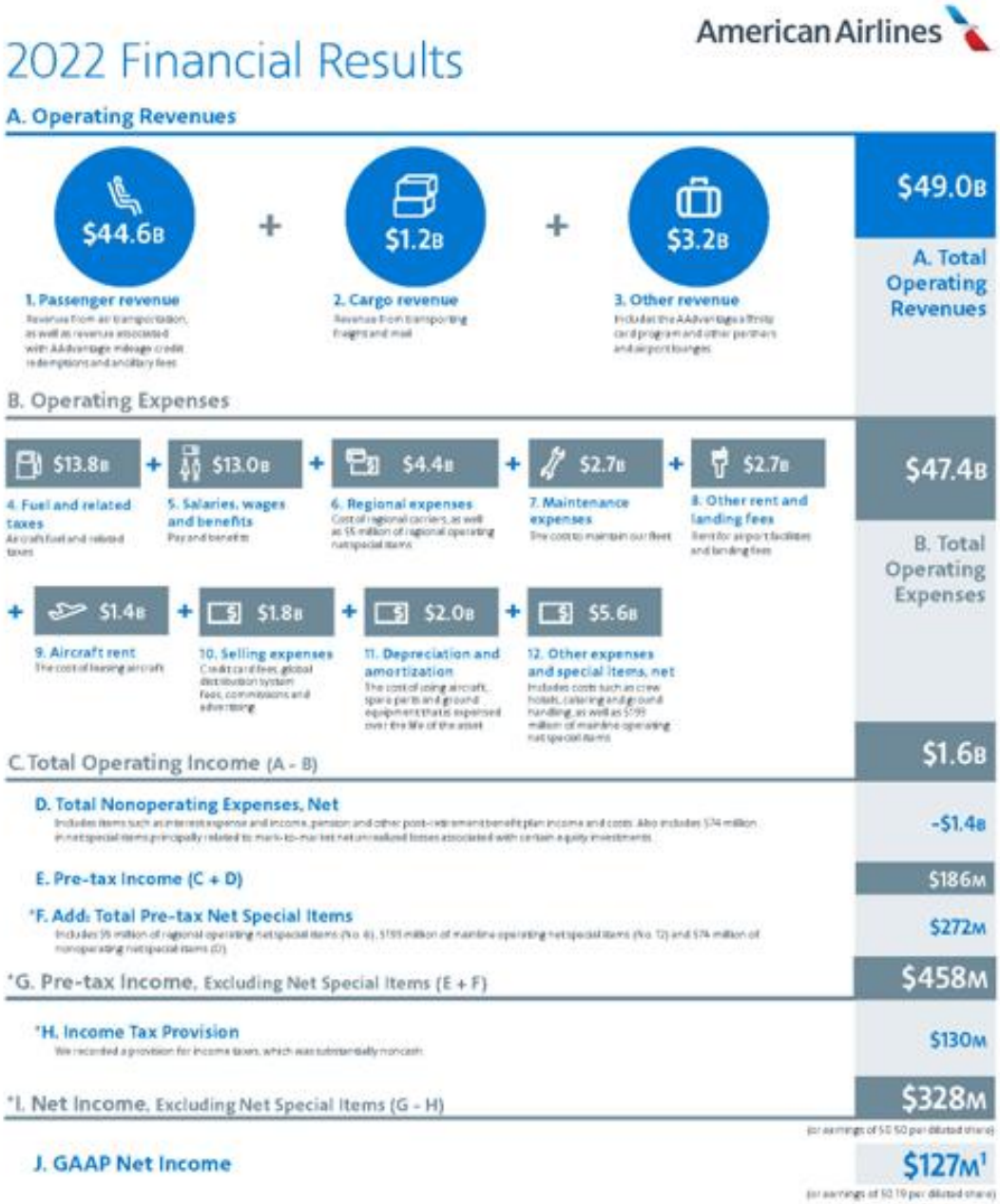
- Growing demand → Grows rapidly
- Needs Significant Capital →Heavy use of Capital.
- Earns no money → ROIC < WACC
- High Fixed Cost Structure →

  Load Factor = nº Passengers / Nº seats available → Use of capacity
- Dependent on oil prices (25% operating costs)
- Deregulation (1979) → Lots of new entrance → Competition Price wars.
- Two business Models: Legacy Airlines vs. Low cost Airlines (new)

INDUSTRY PERFORMANCE (Recent years)

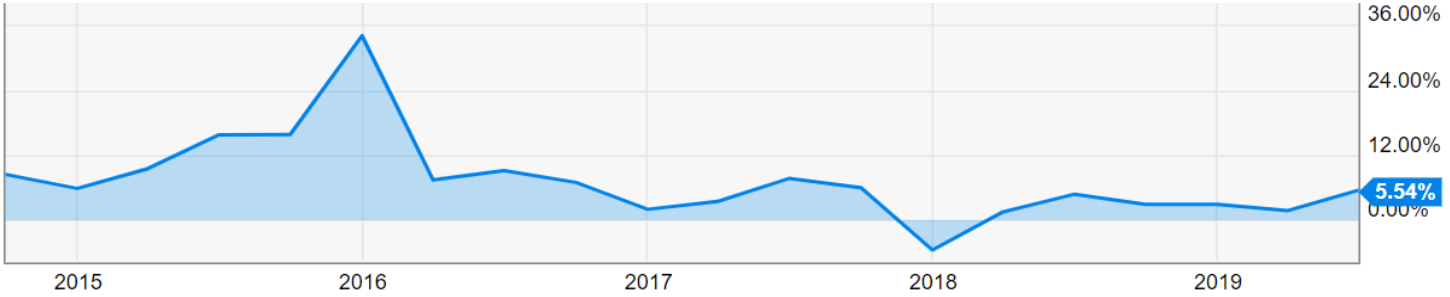
- Increase demand. The economic recovery after the 2008 financial crisis increased demand for air travel, benefiting the industry.
- Lower Oil Prices: A sharp decline in oil prices in 2015 and 2016 helped airlines cut operating costs
- Improved Cost Efficiency: Airlines improved their cost structures through union, concessions, outsourcing, and investments in fuel-efficient aircraft
- Industry Consolidation: Major mergers and acquisitions reduced the number of players.
- Capacity Management: Airlines focused on filling existing capacity rather than increasing it.

How did the industry perform in recent years?

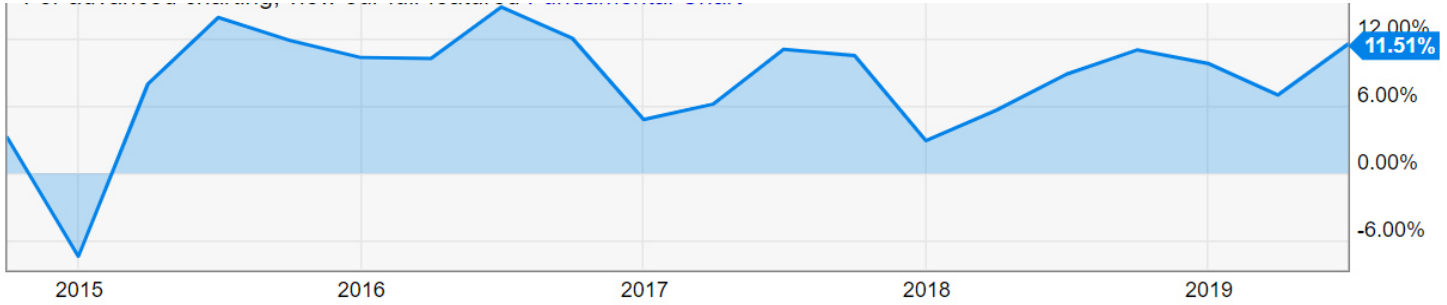


Q1: How did the industry perform in recent years?

American Airlines Profit Margin

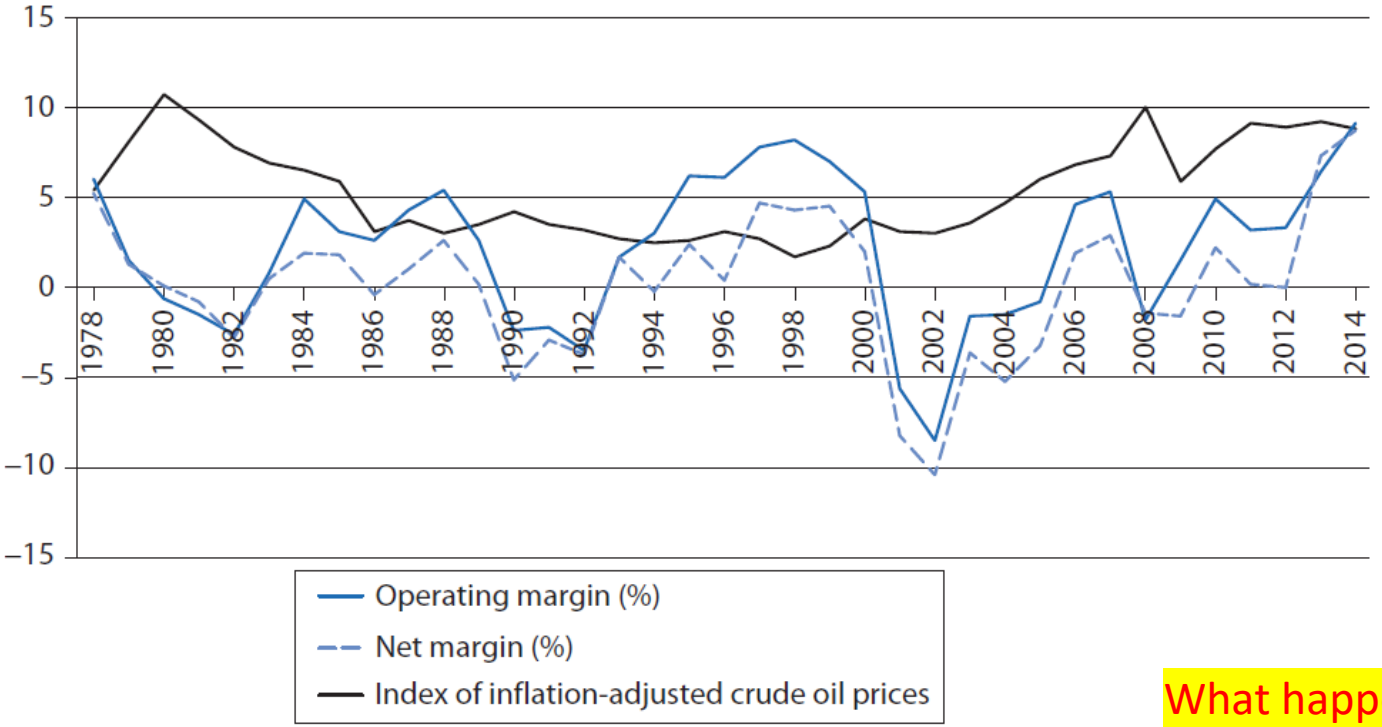


Delta Airlines Profit Margin



Source: ycharts

Q1: How did the industry perform in recent years?

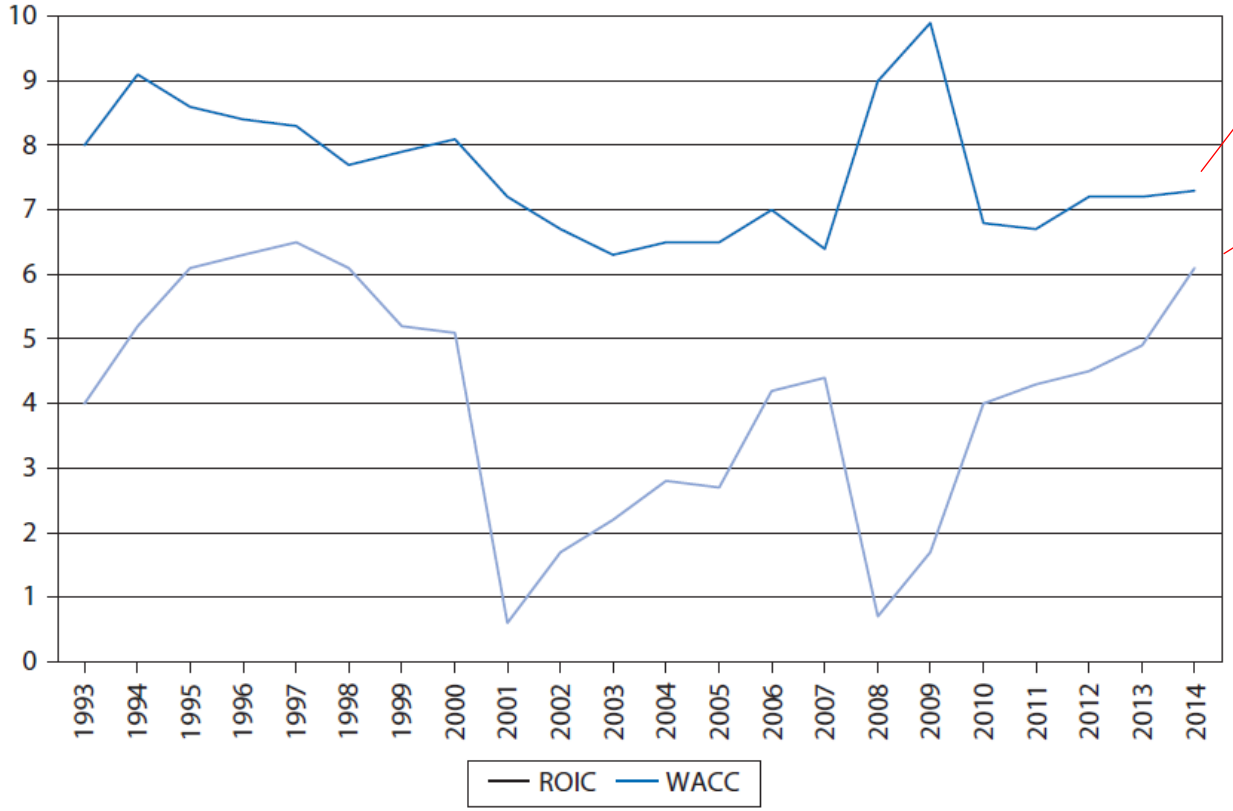


Source: Bureau of Transportation Statistics.

What happen if the inflation index is higher than the net margin?

Q1: How did the industry perform in recent years?

FIGURE 2 Return on invested capital (ROIC) and weighted average cost of capital (WACC) for the world airline industry, 1993–2014



Source: IATA.

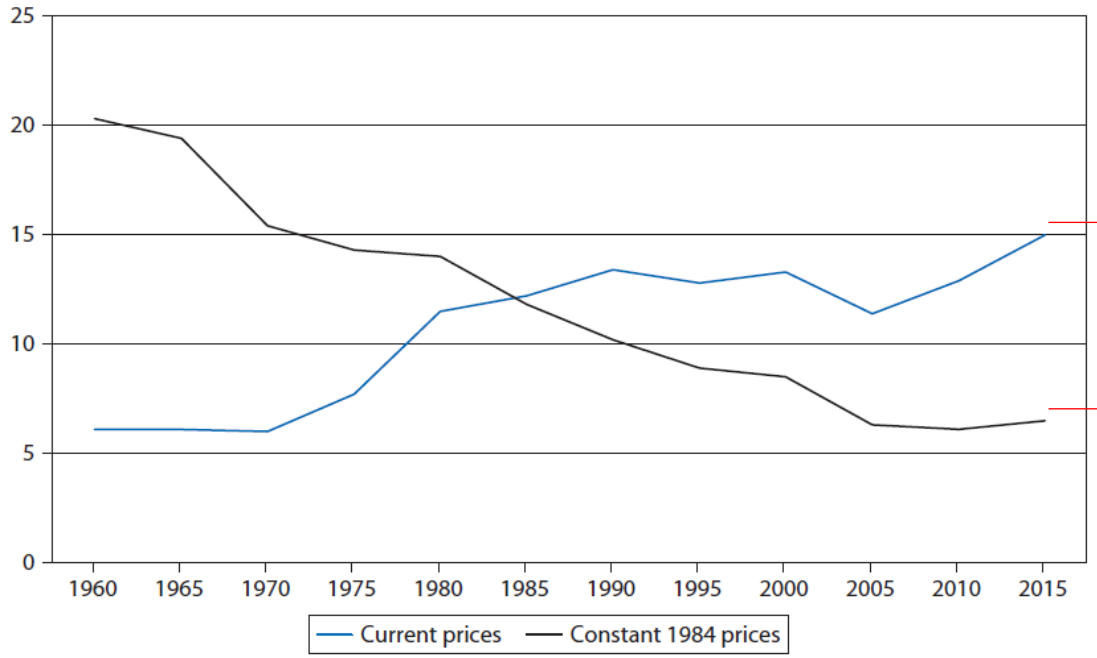
WACC: Weight Average Cost of Capital

ROIC: Return on invested capital

What happen if the WACC is higher than the ROIC?

Q1: How did the industry perform in recent years?

FIGURE 5 Average fares in the US airline industry (cents per revenue passenger mile), 1960–2015



Source: Bureau of Transportation Statistics.

Decreasing tickets Prices

Current ticket prices

Constant ticket prices - adjust the price with effect of inflation

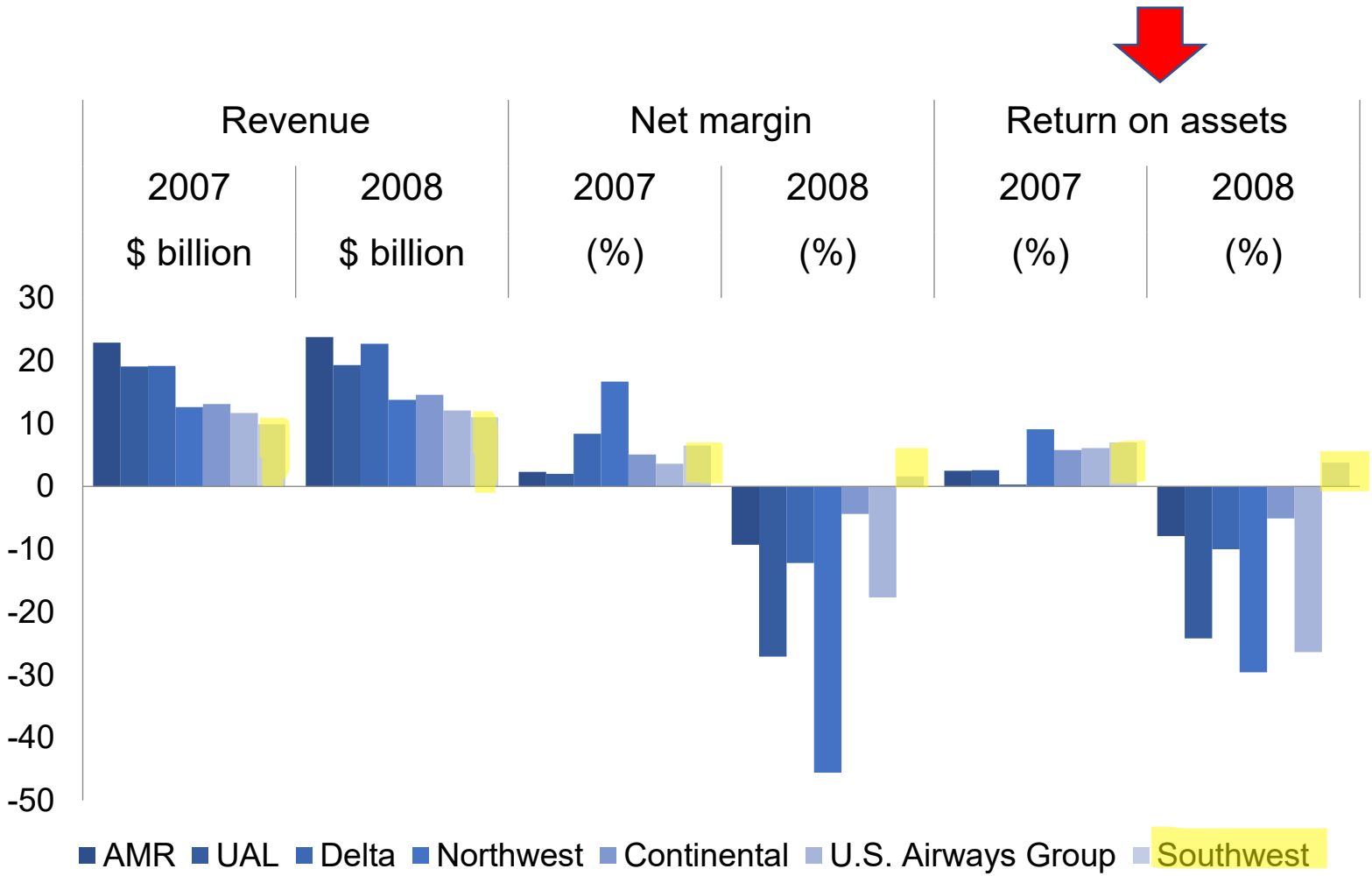
Continuously falling prices in the industry

Q1: How did the industry perform in recent years?

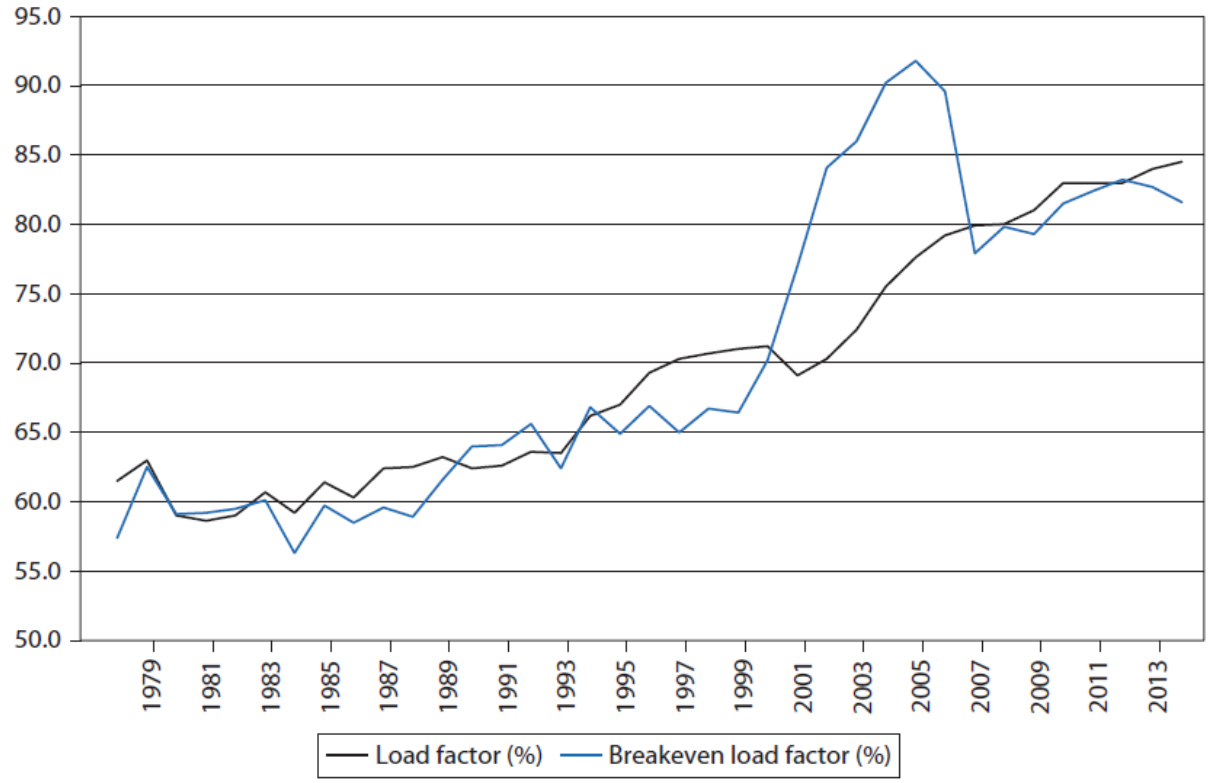
- While having cyclical ups and downs, the industry hasn´t been very profitable in the last decade
- All major companies in the industry have been losing money in some years (exception: Southwest airlines)
- Revenues have increased for each player, but don´t cover costs.



Q1: How did the industry perform in recent years?



Q1: How did the industry perform in recent years?



Source: Air Transport Association, annual economic reports (various years); Bureau of Transportation Statistics.

Increasing Breakeven Load Factor

What happen if the breakeven load factor increases?

What happen if the Load factor is lower than the breakeven load factor?

Passenger load factor = Ratio of revenue passenger miles to available seat miles of a particular transportation operation (e.g. a flight)

- Measures the capacity utilization (assess how efficiently a transport provider fills seats and generates fare revenue)

Q1: How did the industry perform in recent years?

- **In sum:**
  - **Falling prices**
  - **Increasing break even load-factors**
  - **Low profitability**
  - **Not very attractive industry**

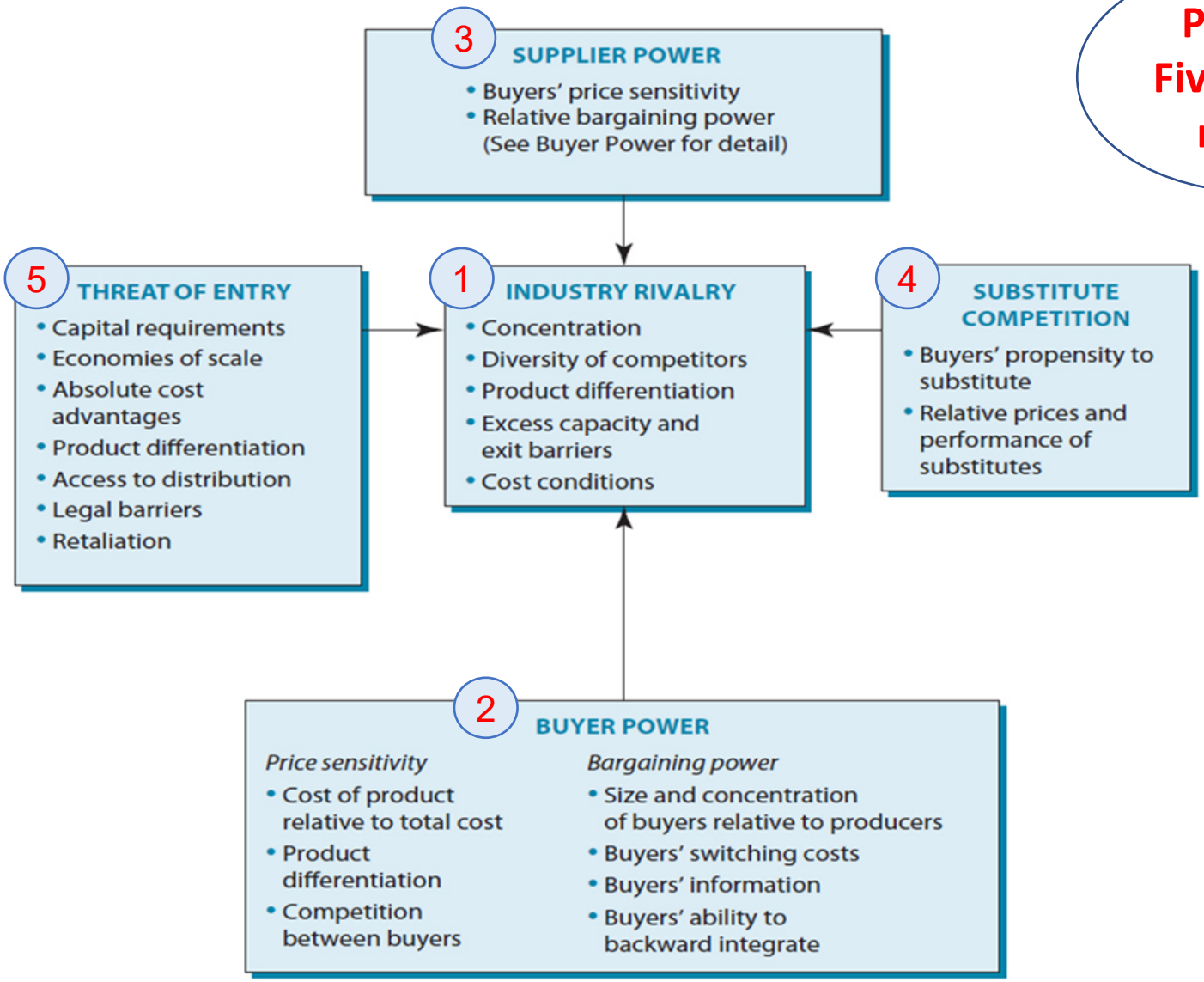
… Why?

Question 2→

Can the low profitability be explained by the industry structure?

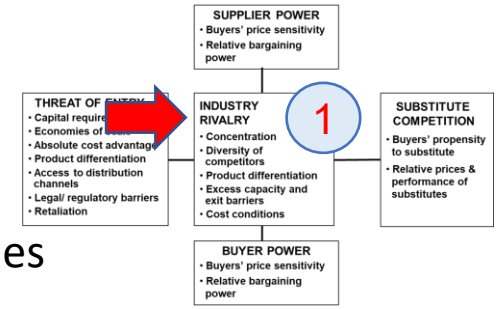
Q2: Can the low profitability be explained by the industry structure?

Porter´s Five Forces model



Rivalry (Intensity of Competition)

Strong rivalry



1. High number of players:
   - This is due to the deregulations and the rise of LCC Airlines

2. Level of differentiation:
   - Product and service not so different
   - Different organizations of routes that impact on the next three determinants

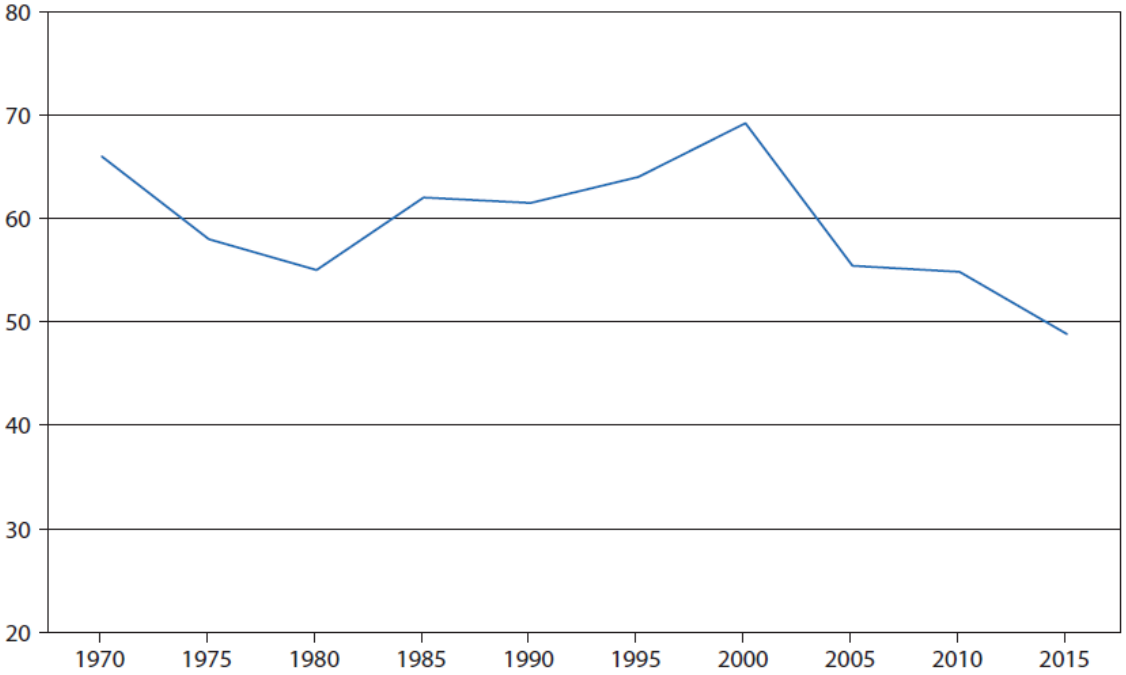
3. High exit barriers:
   - long lived assets + too big to fail

4. Excess capacity:
   - Huge excess capacity to due low load factor

5. High fixed costs:
   - High fixed/variable cost ratio which drives price competition

Rivalry: Industry structure

CR4 Index for US Airline market



Note:
The four-firm concentration ratio (CR4) measures the share of the industry's passenger miles accounted for by the four largest companies. During 1970–1981, the four biggest companies were United, American, TWA, and Eastern. During 1982–2005, the four biggest companies were American, United, Delta, and Northwest. During 2006–2015, the four biggest were American, United, Delta, and Southwest.
Source: US Department of Transportation.

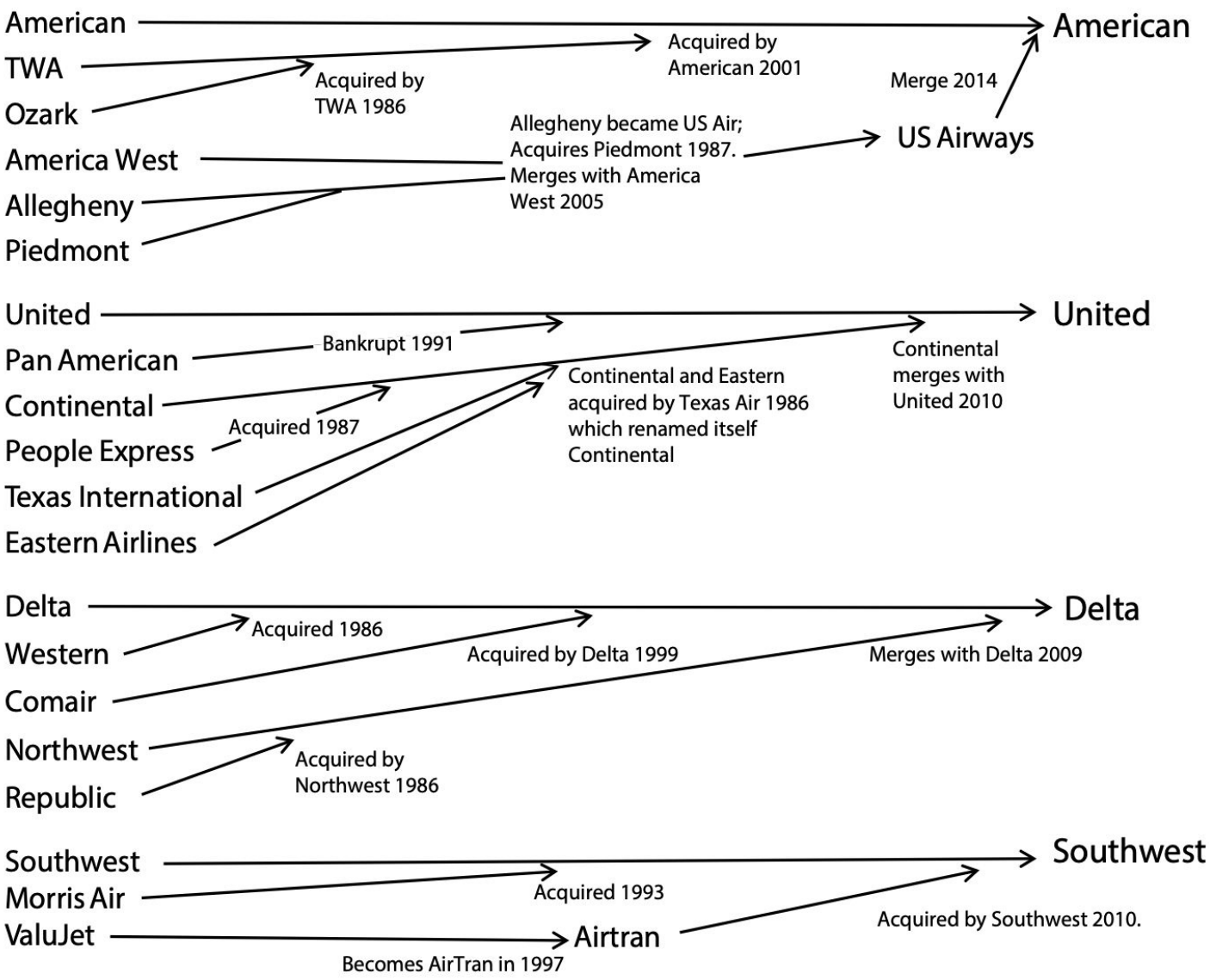
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CR4 Index measures the MARKET SHARE of the biggest 4 competitors in the industry

- Possible alternative measure for CR4 → Herfindahl index

INDUSTRY CONSOLIDATION

1



Cost Structure Average Industry

Are these costs FIXED or VARIABLE? 1

	Increase in cost (%)	Percentage of total operating expenses	
Cost item	**2002–8**	**2006**	**2008**
Labor	−6.0	23.8	24.36
Fuel	327.9	25.5	35.94
Professional services	49.9	7.8	8.07
Food and beverage	−11.4	1.5	1, 25
Landing fees	31.4	2	1.93
Maintenance material	34.6	1.4	2.29
Aircraft insurance	−52.0	0.1	0.48
Passenger commissions	−27.0	1.3	0.97
Communication	4.3	0.9	1.01
Advertising and promotion	2.2	0.8	0.60
Transport-related expenses	167.4	14.7	19.23
Other operating expenses	93.6	7.6	3.86

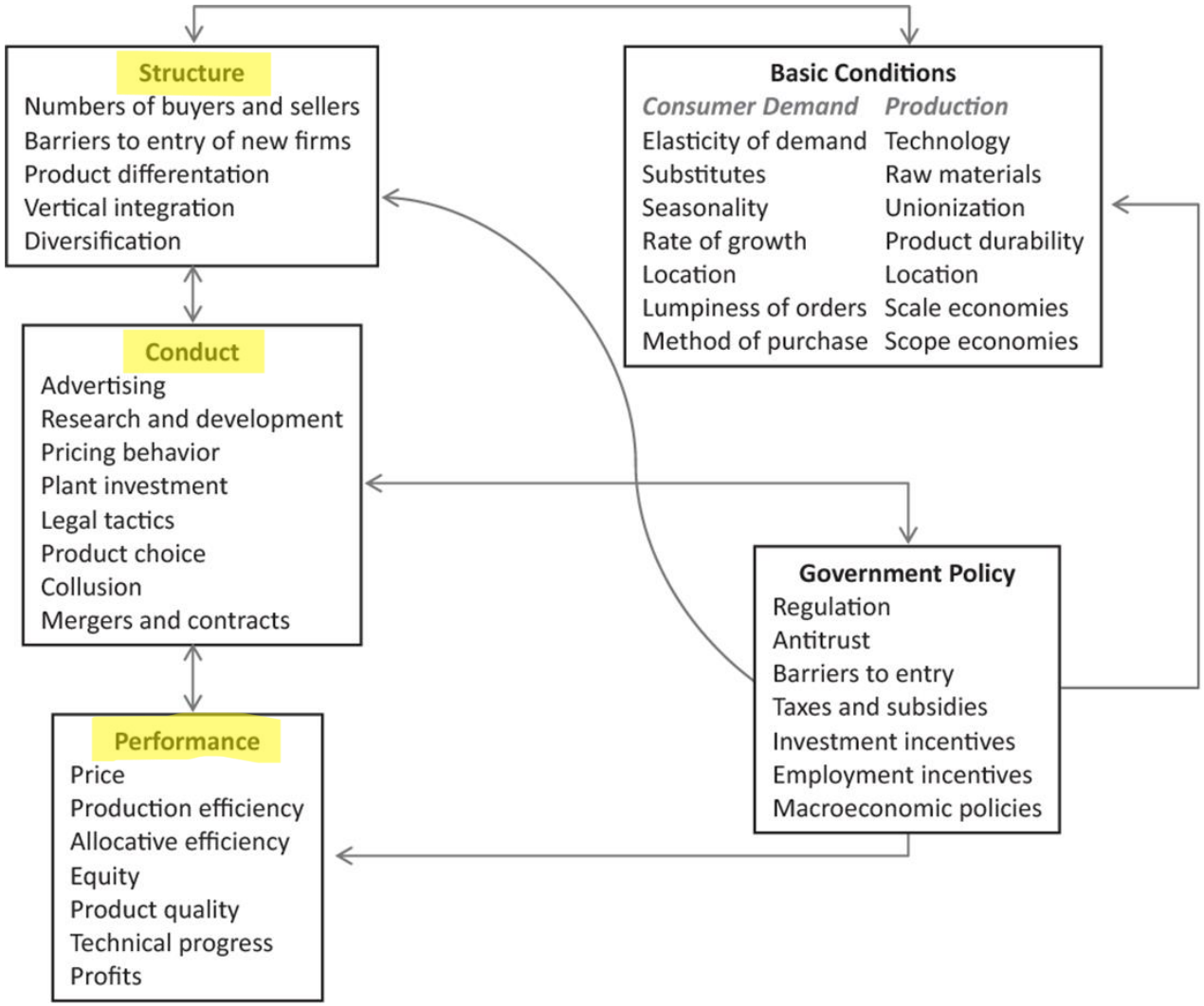
60% to 70%

Airport facilities

- Labor costs were 58% above average US industry
- Fuel was very volatile. Financial options (forward contracts)

How do the Structure (S) of the industry and the Conduct (C) of firms affected the Performance of the industry (P)?

1



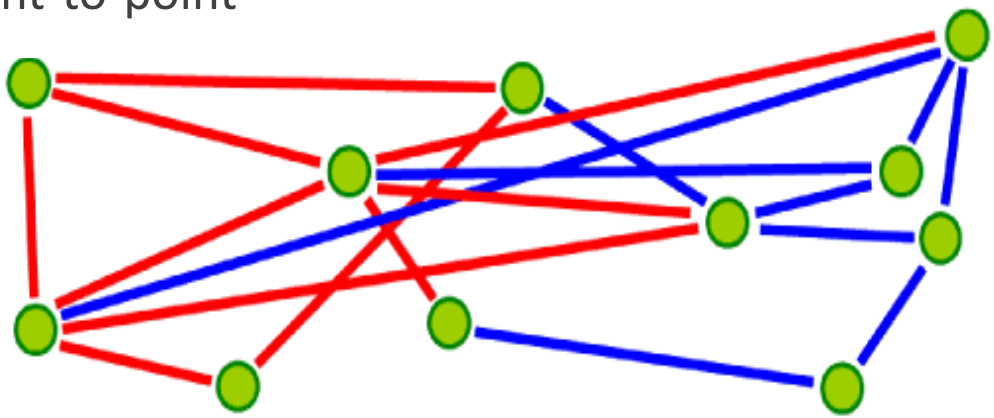
Structure
Pe

How do the Structure (S) of the industry and the Conduct (C) of firms affected the Performance of the industry (P)?

1

- M&As and alliances
  - Increased concentration
  - Shared services (ticketing, pricing, plan servicing) + shared branding among clustered airlines allow to split costs/investments and foster cross-selling (Ex. One world; Star Alliance; …)
- Emergence of Low-cost carriers (LLCs)
  - Increased differentiation
  - Difficult for traditional airlines to adopt the LCC model (requires radically different value chain, culture, combination of routes to maximize efficiency)
- Infrastructural reorganization from Point-to-point to Hub & spoke
  - Increased differentiation (geographical clusters)
  - More efficiency
  - Less redundancy
  - **BUT this had a significant drawback: Airports (Hubs) have increased bargaining power in assigning slots**

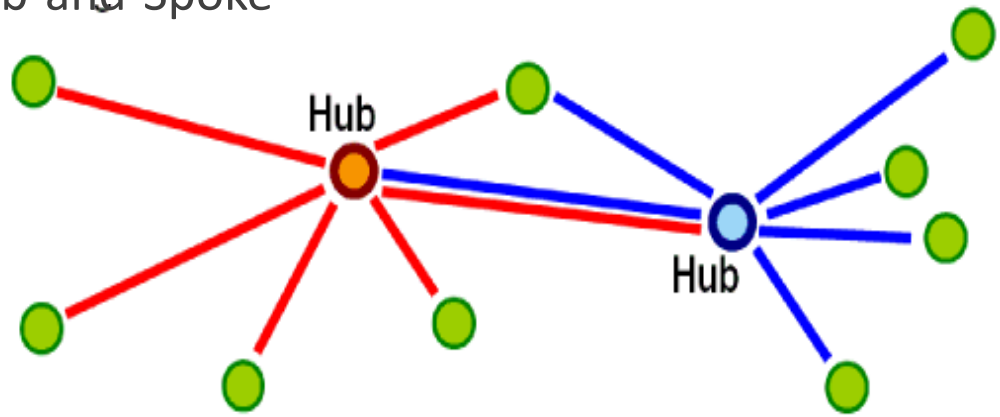
Point-to-point



Rivalry of competitors

SMALL TO MIDDLE AIRCRAFT

Hub-and-Spoke

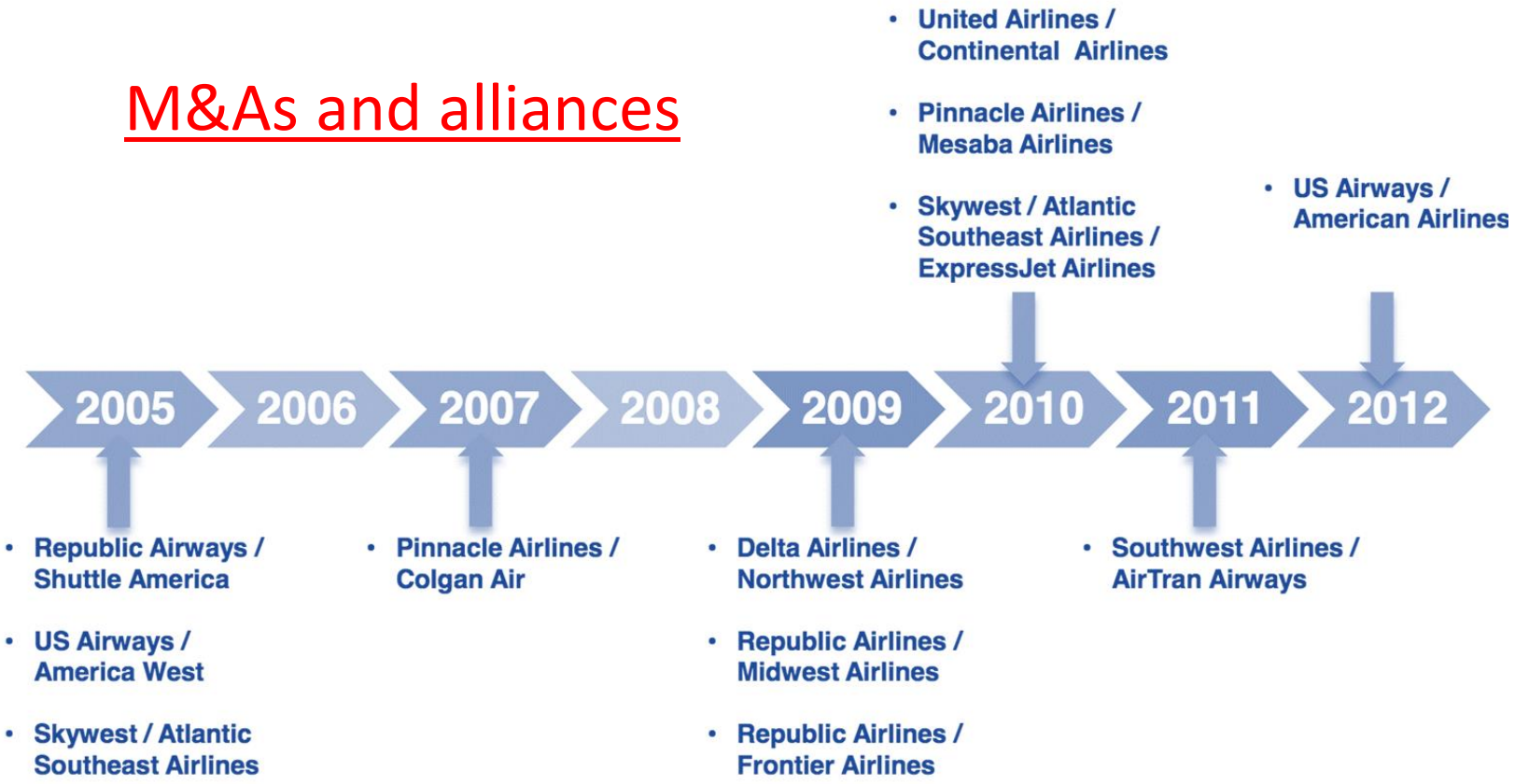


LARGE AIRCRAFT

1

Did rivalry also influence industry structure? (hypercompetition)

M&As and alliances



Differences between LCC vs. Legacy airlines

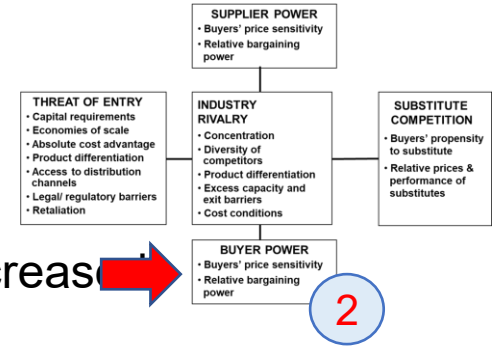
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Diversity of Competitors

- LCC vs. Legacy Airlines:
  - Differences in cost structure :
    - Structural costs
    - Fuel hedging
    - Labor productivity
  - Differences in service
  - LCC more adaptable to change

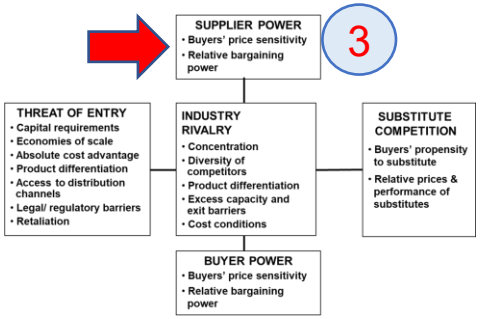
- **CASE: Southwest Airlines: Where does the cost-based advantage come from?**
  - **Southwest Airlines gests more value from customers and employees**

Industry structure: Buyer power



- Medium/ High buyer power Why?:
  - Price sensitivity and access to information have increas[illegible] enormously since tickets are sold online
  - Low switching costs
  - Airlines try to escape the commodity trap by:
    - Price discrimination and yield management, i.e. segmenting their offering by flexibility (early birds, costs for ticket changes, etc.)
    - Customer loyalty programs
  - Still, given the huge competition, all
  - Major cost reductions are passed over to customers

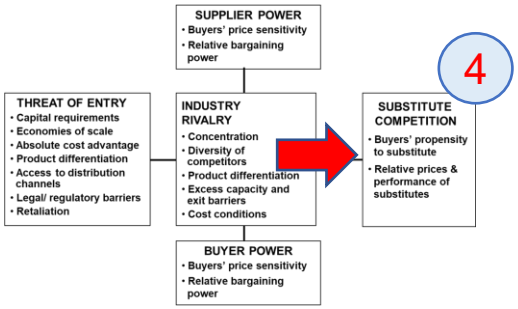
Industry structure: Supplier power



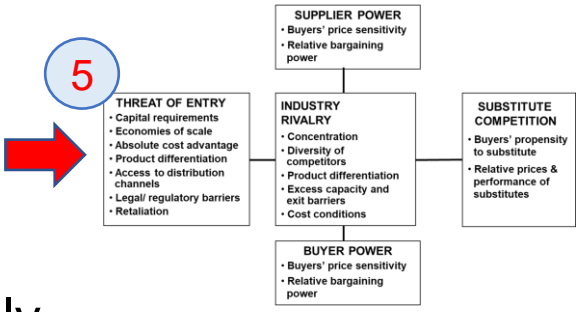
- High power

- Why?:
  - Very few airplane producers (Boeing + Airbus)
  - Strong unions
  - Airports (Hubs) have a local monopoly
  - Few fuel providers, but airlines have become more flexible given *hedging and backward integration*

Substitutes and complements



- Medium threat from substitutes Why?:
- While there are other means of transportation, they only substitute for short-range routes.
- **Videoconferencing** may be considered a strong substitute, and technology has improved and will develop further.
- External → Health threats (COVID), geopolitical threats, wars.
- No strong dependency on complements
  - Alliances with hotels, limousine services etc. are possible
  - Attractive for high-end travelers, but not really for average customer
  - Not a major driver of industry success



Medium threat of entry Why?:

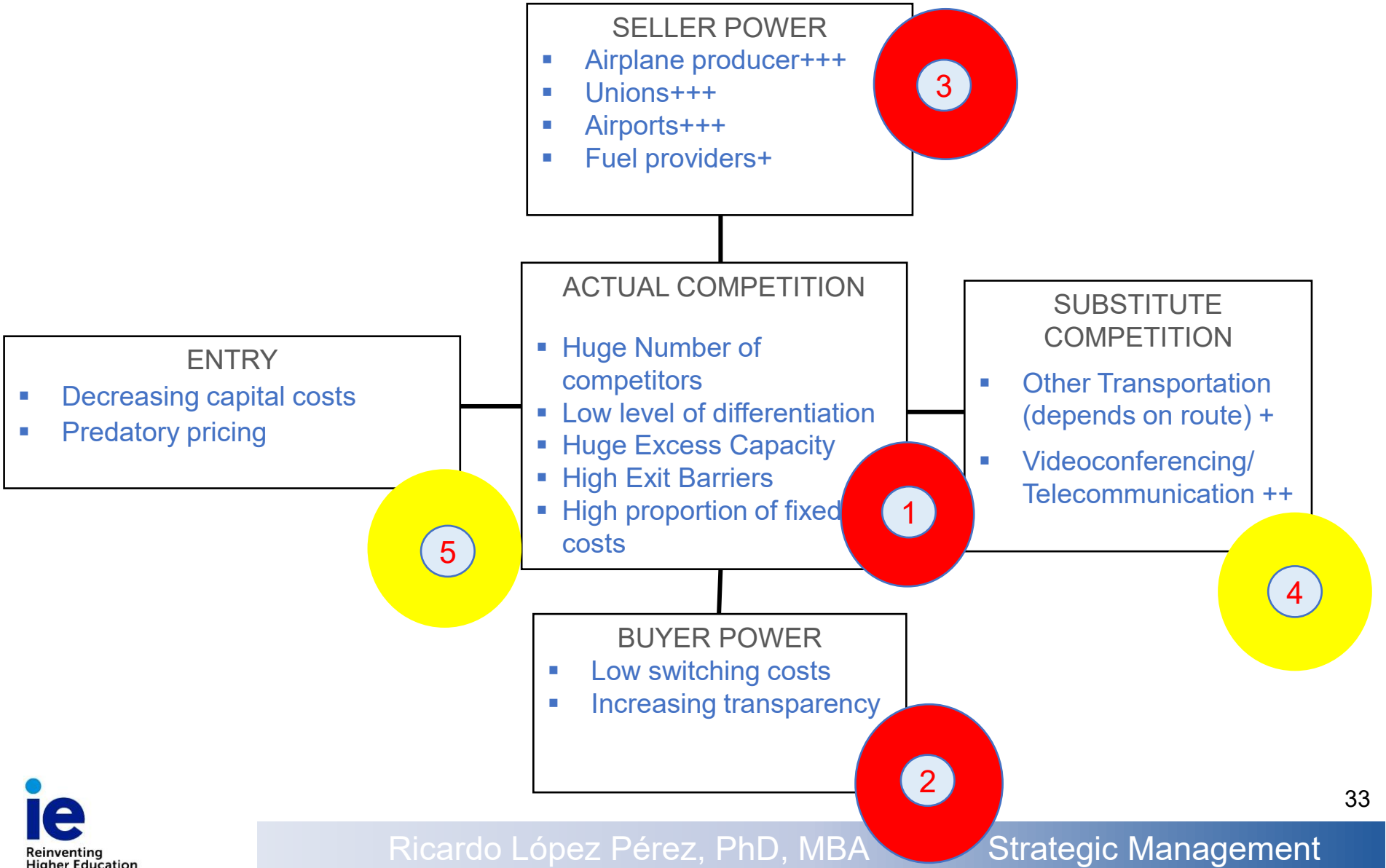
- Barriers to entry have become lower, especially because capital requirements have decreased
  - Most assets can be leased
- Is the industry attractive?
  - Besides the low barriers to entry,
  - it is unclear why entrepreneurs keep on entering the industry?
  - Behavioral explanations….



Question 3→

Which of Porter's five forces has had the biggest impact in depressing industry profitability?

Can the low profitability be explained by the industry structure?



Summary

- Not attractive
- High fixed costs and unfavorable cost structure
  - High proportion of fixed costs
- Little room for differentiation strategy
- Moderate buyer and high supplier power
- Unattractive for incumbents and new entrants
- The central problem is competition – firms are selling undifferentiated products and compete on price -> costs; this is further exacerbated by high supplier power and moderate buyer power

Question 4→

How will industry profitability develop during the next five years? What would you do as manager of an airline?

How will industry profitability develop during the next five years?

What would you do as manager of an airline?

- Differentiation
  - Extra services (free Wi-Fi)
  - Forward integration
- Development of adjustment of capacity relative to demand
  - Risk of overcapacity
- M&As
  - Need to be allowed by competition authorities
  - Difficult to manage integration
- Cost reductions
  - Likely to benefit consumers more than the firms
- Entry of Asia competitors
  - Important to keep an eye on eventual new entrants and overseas carriers
- Supplier power
  - Unlikely to change

Identifying KSFs Through Modeling Profitability: The Airline Industry

Profitability = Yield x Load factor - Unit Cost

$$\frac{\text{Income}}{\text{ASMs}} = \frac{\text{Revenue}}{\text{RPMs}} \times \frac{\text{RPMs}}{\text{ASMs}} - \text{Expenses}$$

- Strength of competition on routes.
- Responsiveness to changing market conditions
- % business travelers.
- Achieving differentiation advantage

- Price competitiveness.
- Efficiency of route planning.
- Flexibility and responsiveness.
- Customer loyalty.
- Meeting customer requirements.

- Wage rates.
- Fuel efficiency of planes.
- Employee productivity.
- Load factors.
- Administrative overhead.

NOTE: **ASM = Available Seat Miles; RPM = Revenue Passenger Miles**

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Key take-aways for the next case analysis

- Porter Five Forces can help
  - To identify the key structural factors that drive competition
  - To predict trends for the intensity of competition and levels of profitability
  - To suggest strategies that can moderate the forces of competition in order to increase overall industry attractiveness
- Porter Five Forces analysis (like most other strategy frameworks) is not an algorithm that provides clear and precise answers.

Additional Information of the Industry

FLAGSHIP AIRLINE BUSINESS MODEL

The Rise and Fall of American Airlines (15 mts.)
https://www.youtube.com/watch?v=W6sQBmEx-s4



LOW COST AIRLINE BUSINESS MODEL

Southwest Airlines - Case Study | Success Story | Strategy (9mts.)
https://www.youtube.com/watch?v=qILkq-raqXs



Identifying Key Success Factors by Profitability Modeling: Airlines

Profitability, as measured by operating income per available seat-mile (ASM), is determined by three factors: yield, which is total operating revenues divided by the number of revenue passenger miles (RPMs); load factor, which is the ratio of RPMs to ASMs; and unit cost, which is total operating expenses divided by ASMs. Thus:

$$\frac{\text{Profit}}{\text{ASMs}} = \frac{\text{Revenue}}{\text{RPMs}} \times \frac{\text{RPMs}}{\text{ASMs}} - \frac{\text{Expenses}}{\text{ASMs}}$$

Some of the main determinants of each of these component ratios are the following:

- Revenue/RPMs
  - intensity of competition on routes flown
  - effective yield management to permit quick price adjustment to changing market conditions
  - ability to attract business customers
  - superior customer service.
- Load factor (RPMs/ASMs)
  - competitiveness and flexibility of prices
  - efficiency of route planning (e.g., through hub-and-spoke systems)
  - building customer loyalty through quality of service, frequent-flier programs
  - matching airplane size to demand for individual flights.
- Expenses/ASMs
  - wage rates and benefit levels
  - fuel efficiency of aircraft
  - productivity of employees (determined partly by their job flexibility)
  - load factors
  - level of administrative cost.

Thanks a lot!

Any questions?

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